

Hardware Engineering Chapter

Open Gauge: An Open-Source, Dual-Node Wireless Vehicle Telemetry and Secondary Instrument Gauge Platform

This chapter presents the complete hardware engineering specification for Open Gauge, covering system-level design rationale, the OBD-II acquisition dongle (Board 1), the cockpit display node (Board 2), the consolidated bill of materials, and the manufacturing and PCB layout guidelines used to realize the design.

1. System-Level Design Rationale

1.1 Asymmetric Dual-Node Topology

Open Gauge partitions functionality across two physically separate, wirelessly linked printed circuit boards rather than adopting the conventional single-board approach used by most hobbyist OBD-II gauge projects. Board 1 (the Acquisition Dongle) resides permanently inside the OBD-II tunnel, exposed to the harshest electrical environment in the vehicle, while Board 2 (the Display Node) is a battery-backed, purely-digital cockpit instrument mounted for driver visibility. This asymmetry is a deliberate engineering decision, not an artifact of convenience, and is justified along four axes below.

1.1.1 Electrical isolation of the hostile domain

A traditional single-board wired gauge cluster must route both the raw vehicle power/CAN domain and the low-noise display/touch domain on one substrate, forcing the sensitive SPI and I2C traces feeding the display controller to share a ground plane with a switching buck converter fed directly from an unregulated automotive rail. By splitting the boards, all automotive-grade transient handling (load dump, reverse polarity, ESD on the CAN lines) is physically confined to Board 1, and the only signal crossing the boundary to Board 2 is a 2.4 GHz radio packet — there is no galvanic connection between the vehicle electrical system and the cockpit-mounted display at all.

1.1.2 Independent power budgets and duty cycles

Board 1 is permanently powered whenever the vehicle battery is connected (or gated by switched 12 V, depending on installation), and its power budget is dominated by the CAN transceiver and RF broadcast duty cycle. Board 2 is a battery-powered, user-facing instrument that must survive extended operation off a small LiPo cell when removed from USB power, and its budget is dominated by the TFT backlight and the LVGL render pipeline. Decoupling the two boards allows each power subsystem to be optimized independently: Board 1 needs a wide-input automotive buck regulator, while Board 2 needs an efficient linear power-path with instant source switchover rather than a second automotive-rated buck stage it would never use in normal operation.

1.1.3 Mechanical placement freedom

The OBD-II port is fixed by the vehicle manufacturer under the steering column, an ergonomically poor location for a driver-facing gauge. A single-board design would force an unsightly cable run from the port to the dash. The dual-node topology lets Board 1 remain hidden at the port while Board 2 is placed anywhere convenient — vent-mounted, bezel-mounted, or handheld — with ESP-NOW eliminating the wiring harness entirely.

1.1.4 Fault containment and field serviceability

Because the boards are only coupled over RF, a fault on one node (e.g. a CAN transceiver damaged by an out-of-spec transient) cannot propagate to or brick the other. Each board can be reflashed, replaced, or bench-tested independently over its native USB interface without disturbing the other node, which materially reduces field-service complexity relative to a monolithic board where a single fault domain covers both acquisition and display electronics.

1.2 Automotive Power Rail Dynamics

Board 1 draws power directly from the vehicle's 12 V nominal rail at the OBD-II connector, a supply that is nominal in name only. The regulator, input protection network, and PCB layout were all sized against the transient environment defined by ISO 7637-2 and observed alternator/starter behavior, summarized below.

Transient class	Origin	Typical magnitude
Pulse 1	Supply line disconnect while inductive loads are switched	–100 V (clamped), fast decay

Transient class	Origin	Typical magnitude
Pulse 2a	Supply interruption due to voltage drop at wiring resistance	+75 V transient, short duration
Pulse 3a/3b	Transients from switching processes on the bus wiring	±150 V, sub-microsecond, fast repetitive
Pulse 5b (load dump)	Alternator field-decay after battery disconnect under charge	+40 V typical, several hundred ms, up to seconds
Cold crank	Starter motor draw collapsing battery voltage	Down to 6 V for ~15–20 ms, ringing recovery
Alternator ripple	Rectifier ripple superimposed on the charging rail	±0.5–1 V at 3× to 6× engine RPM harmonic

Table 1.1 — ISO 7637-2 conducted transient classes relevant to the Board 1 input stage.

The design response to this environment is layered rather than relying on a single component: the TPS54302 buck regulator is specified for a 4.5–28 V continuous input range, comfortably spanning cold-crank sag (6 V) through nominal charging voltage (13.8–14.4 V); a SMAJ24A/SMBJ28A unidirectional TVS diode clamps load-dump energy at the input before it reaches the regulator’s absolute maximum rating; and the SS34 series Schottky diode blocks reverse-polarity connection at the connector, which — unlike the transient pulses above — is a static fault condition rather than a momentary event and must be tolerated indefinitely without damage. Alternator ripple, being a relatively low-amplitude, low-frequency artifact well within the buck converter’s input operating range and further attenuated by the input bulk capacitance, does not require dedicated suppression beyond the standard input filter network.

2. Board 1: OBD-II Telemetry Acquisition Dongle

2.1 Core Processing: ESP32-C3-WROOM-02

Board 1 is built around the Espressif ESP32-C3-WROOM-02-N4 castellated SMT module, integrating a 32-bit single-core RISC-V processor, 4 MB of SPI flash inside an RF shielding can, and the on-chip TWAI (Two-Wire Automotive Interface) controller that implements the CAN 2.0B protocol engine used for OBD-II polling.

The core clock is derived from a 40 MHz external crystal reference through the chip's internal phase-locked loop, which multiplies the reference up to the processor's 160 MHz operating frequency:

$$f_{CPU} = f_{XTAL} \times N_{PLL} = 40 \text{ MHz} \times 4 = 160 \text{ MHz}$$

This single-core budget is sufficient because Board 1's workload is I/O-bound rather than compute-bound: the dominant task is scheduling periodic CAN request frames, parsing single-frame ISO 15765-4 responses, and packing the result into a fixed 10-byte ESP-NOW payload, none of which approaches the processor's throughput ceiling even at the fast-polling group's 10–20 Hz cadence.

2.2 Power Stage Analysis — TPS54302 Synchronous Buck

The TPS54302 is a synchronous step-down converter with integrated high-side and low-side FETs, selected for its wide 4.5–28 V input range (matching the transient envelope described in §1.2), integrated compensation, and small external component count. The design targets a fixed 3.3 V output at up to 1.5–2.0 A, switching at approximately 400–500 kHz.

2.2.1 Output inductor selection

The output inductor is sized from the target peak-to-peak ripple current, expressed as a fraction K_{IND} of the maximum DC load current (a design value of $K_{IND} = 0.3 - 0.4$ is standard practice for this class of automotive buck converter, balancing inductor size against ripple-induced core and conduction losses):

$$L1 = [V_{OUT} \times (V_{IN(max)} - V_{OUT})] / [V_{IN(max)} \times f_{SW} \times \Delta I_L]$$

where $\Delta I_L = K_{IND} \times I_{OUT(max)}$. Substituting a worst-case $V_{IN(max)}$ of 16 V (clamped alternator ceiling), $V_{OUT} = 3.3$ V, $f_{SW} = 450$ kHz, $I_{OUT(max)} = 1.5$ A and $K_{IND} = 0.35$ gives $\Delta I_L \approx 0.525$ A and yields an inductor value in the 6.5–7.5 μ H range — consistent with the specified 4.7–10 μ H shielded power inductor with a saturation current rating $I_{SAT} \geq 2.0$ A, which retains adequate margin above the calculated peak current ($I_{OUT(max)} + \Delta I_L / 2 \approx 1.76$ A).

2.2.2 Input capacitor ESR and RMS current

The buck converter draws pulsed current from the input rail at the switching frequency; the input capacitor must supply this ripple current locally so that only DC (and low-frequency alternator ripple, already discussed) is drawn from the harness. The input capacitor RMS current requirement is approximated by:

$$I_{CIN(RMS)} \approx I_{OUT} \times \sqrt{[D \times (1 - D)]}$$

where $D = V_{OUT} / V_{IN}$ is the converter duty cycle. At the nominal 13.8 V input, $D \approx 0.24$, giving $I_{CIN(RMS)} \approx 0.65 \times I_{OUT}$. The specified 10 μ F 50 V X7R ceramic input capacitor, combined with a 100 nF high-frequency bypass placed immediately at the IC's VIN pin, presents ESR in the low tens of milliohms — low enough that the resulting input ripple voltage ($\Delta V_{IN} \approx I_{CIN(RMS)} \times \text{ESR}$) stays a small fraction of a volt, well within the regulator's tolerance for line noise.

2.2.3 Feedback divider for fixed 3.3 V output

For adjustable-output variants of this converter family, the output voltage is programmed by a resistive divider from V_{OUT} to the feedback pin, referenced against the internal bandgap V_{REF} (typically 0.8 V):

$$V_{OUT} = V_{REF} \times (1 + R1/R2)$$

Rearranged for a fixed 3.3 V target with a standard $R2 = 10.0 \text{ k}\Omega$ (1% tolerance) lower divider resistor:

$$R1 = R2 \times (V_{OUT}/V_{REF} - 1) = 10.0 \text{ k}\Omega \times (3.3/0.8 - 1) \approx 31.25 \text{ k}\Omega$$

The nearest E96 1% value (31.6 k Ω) is used in practice, giving an actual output of approximately 3.328 V — within the $\pm 2\%$ tolerance the downstream digital logic and CAN transceiver both accept.

2.3 Physical CAN Bus Interface — SN65HVD230

The SN65HVD230DR is a 3.3 V-native CAN transceiver compliant with ISO 11898-2, chosen specifically because it eliminates the level-shifting network that would otherwise be required between a 3.3 V MCU and a 5 V-logic transceiver, reducing both component count and an additional failure surface on the bus-facing side of the board.

TVS protection is provided by a NUP2105L dual-line automotive ESD/EFT suppressor placed directly across CAN_H and CAN_L, as close to the J1962 connector as routing allows, so that transient energy is diverted before it reaches the transceiver's bus pins.

The 120 Ω termination footprint (JP1) is deliberately left open by default. Vehicle CAN buses already present the correct differential termination (typically two 120 Ω terminators at the ECU and gateway ends of the bus, combining to a nominal 60 Ω differential load); adding a third 120 Ω termination at the dongle would over-load the bus, degrading signal integrity rather than improving it. JP1 exists only to permit standalone bench testing of Board 1 against a CAN bus simulator, where no other termination is present.

2.4 Dual-Power Conflict Prevention

During bench calibration it is common to have Board 1 connected simultaneously to a laptop (via USB-C, for flashing/logging) and to a bench 12 V supply or live vehicle harness (for CAN traffic). Without protection, the buck converter's 3.3 V output — or in a fault condition, the raw 12 V rail — could back-feed through the USB connector into the laptop's USB port. A low forward-voltage Schottky diode (BAT54 or SS14) is placed in series with the USB-C 5 V VBUS net, oriented so that current can only flow from the USB host into the board, never the reverse direction, isolating the two supply domains during concurrent connection.

3. Board 2: Cockpit Instrument Display Node

3.1 Processing Core — ESP32-S3-WROOM-1-N8R8

Board 2 uses the dual-core Xtensa LX7 ESP32-S3-WROOM-1-N8R8 module, clocked at 240 MHz from the same 40 MHz-reference PLL architecture used on Board 1, with 512 kB internal SRAM augmented by 8 MB of embedded Octal PSRAM and 8 MB of Quad SPI flash. The additional core (relative to Board 1's single-core part) and the PSRAM frame buffer are both required by the graphics workload rather than by the wireless link.

Task allocation is fixed by design: Core 0 owns the wireless protocol stack — unpacking incoming ESP-NOW telemetry frames and servicing background BLE tasks — while Core 1 is dedicated exclusively to the LVGL rendering pipeline and the DMA-driven SPI transfer to the display controller. This separation ensures that radio interrupt servicing never stalls the frame render loop, and that a full 240×240 RGB565 frame buffer (240 × 240 × 2 bytes = 115,200 bytes per frame) can be held in PSRAM without contending with the 512 kB internal SRAM budget needed for the wireless stack and LVGL object heap.

3.2 Display & Touch Subsystem

The panel is a Waveshare 1.28-inch circular IPS TFT (240×240, RGB565, active diameter ≈ 32.4 mm) driven by the GC9A01 controller over a 4-wire SPI interface (MOSI, SCLK, CS, DC) with DMA-assisted transfers, paired with a CST816S capacitive touch controller on a standard I2C bus.

3.2.1 SPI throughput and frame-rate budget

Each full-frame transfer moves the entire 115,200-byte RGB565 buffer across the SPI bus. At a 40 MHz SPI clock, the raw bit rate available for data transfer is:

$$R_{bit} = f_{SPI} = 40 \times 10^6 \text{ bit/s} \Rightarrow R_{byte} = 5 \times 10^6 \text{ byte/s (ideal, 8 bit/clock)}$$

The minimum time to shift out one full frame is therefore:

$$t_{frame} = \text{Bytes}_{frame} / R_{byte} = 115,200 / 5,000,000 \approx 23.0 \text{ ms}$$

which corresponds to a theoretical ceiling of roughly 43 FPS for full-frame refresh. In practice LVGL is configured to redraw only the dirty rectangles that changed between frames (needle sweep, changing digits) rather than the full 240×240 buffer every cycle; for a typical dirty-region update covering roughly half the panel area, the transfer time roughly halves to ≈ 11.5 ms, and combined with the DMA engine offloading the transfer from the CPU (allowing LVGL to begin composing the next frame while the previous DMA transfer is still in flight), the pipeline comfortably sustains greater than 30 FPS, satisfying the design requirement.

3.3 USB Input Protection

The USB-C input on Board 2 is protected by a three-stage network: an 0805 surface-mount PTC resettable fuse (500 mA hold / 1 A trip) provides recoverable over-current protection ahead of the charger and system load; a SMAJ5.0A unidirectional TVS diode clamps VBUS transients above approximately 5.5 V; and a USBLC6-2SC6 low-capacitance ESD array protects the D+/D- native-USB data lines without introducing enough parasitic capacitance to distort full-speed USB signal integrity.

3.4 Dynamic Hardware Power-Path & Battery Subsystem

Board 2 must transition seamlessly between USB power and a single-cell 3.7 V LiPo battery (500–1000 mAh) without a software-mediated switchover, since a firmware fault or reset must never leave the display node unpowered mid-transition. This is achieved with a purely analog power-path built from a TP4056 (or MCP73831) linear charger, an AO3401A P-channel MOSFET, and a small Schottky diode network.

3.4.1 Switching dynamics — USB plug-in event

- 1 VBUS rises to 5 V; the TP4056 detects input power and begins charging the LiPo cell (current limited by R_{PROG} , configured for ≈ 400 mA).
- 2 The same VBUS rail is applied to the gate of the AO3401A P-channel MOSFET through the gate network; because V_{GS} becomes strongly positive relative to the battery-referenced source, the P-FET turns OFF, disconnecting the battery from the system rail.
- 3 System load current is now supplied from VBUS through the SS14/BAT54C Schottky diode directly to the system rail, in parallel with (but electrically isolated from) the charging path into the battery.

3.4.2 Switching dynamics — USB unplug event

- 1 VBUS collapses toward 0 V; the TP4056 charge path ceases sourcing current.
- 2 The 100 k Ω pulldown resistor on the P-FET gate, no longer overridden by VBUS, pulls the gate toward the battery-referenced source potential.
- 3 As V_{GS} crosses the AO3401A's threshold, the P-FET turns ON, presenting an ultra-low on-resistance path (< 30 mV drop at typical system load) directly from the battery to the system rail.
- 4 The transition is continuous and sub-millisecond: because the P-FET body diode is oriented to conduct from battery to system rail even before the channel fully turns on, there is no interruption of supply to the ME6211C33 LDO input during the handover, and no visible flicker or reset on the display.

3.5 LDO Regulation — ME6211C33

The ME6211C33M5G (SOT-23-5) provides the final regulated 3.3 V system rail from either the USB-derived or battery-derived unregulated input, with a rated output current of 500 mA and a low dropout voltage of approximately 100 mV at 200 mA load. Because the LiPo discharge curve sags toward roughly 3.4 V near end-of-discharge, the LDO's dropout margin — 3.4 V input minus 3.3 V output leaves only 100 mV of headroom at the low end of the curve — is the limiting factor that defines the battery's usable depth of discharge before the display node begins to brown out; this is consistent with, and was a driving constraint on, the LDO part selection.

4. Complete Bill of Materials

The following tables consolidate every functional component referenced in §2 and §3. Passive component tolerances are 1% for feedback/sense resistors and 10% (X7R or better) for ceramic capacitors unless otherwise noted.

4.1 Board 1 — OBD-II Acquisition Dongle BOM

Ref	Value / Part	Package	Recommended P/N	Function
U1	ESP32-C3-WROOM-02-N4	SMD Module	Espressif ESP32-C3-WROOM-02-N4	MCU / RISC-V / CAN(TWAI) / RF
U2	TPS54302DDCR	SOT-23-6 (Thin)	TI TPS54302DDCR	3.3V synchronous buck regulator
U3	SN65HVD230DR	SOIC-8	TI SN65HVD230DR	CAN 2.0B / ISO 11898-2 transceiver
D1	SS34	SMA / DO-214AC	Diodes Inc. SS34	Reverse-polarity protection
D2	SMAJ24A	SMA / DO-214AC	Littelfuse SMAJ24A	Load-dump / OV TVS clamp
D3	NUP2105L	SOT-23	onsemi NUP2105L	CAN_H/CAN_L ESD/EFT suppressor
D4	BAT54 / SS14	SOT-23 / SMA	Diodes Inc. BAT54	USB back-feed block diode
L1	6.8 uH shielded inductor	4x4 mm SMD shielded	TDK / Wurth SD-series, Isat>=2.0A	Buck output inductor
C1	10 uF 50V X7R	1210	Murata GRM32 series	Buck input bulk capacitor
C2	100 nF 50V X7R	0603	Generic MLCC	Buck input HF bypass
C3,C4	22 uF 10V X7R	0805	Murata GRM21 series	Buck output filter (x2)
R1	31.6 k ohm 1%	0402	Generic thin-film	FB divider upper (V_OUT set)
R2	10.0 k ohm 1%	0402	Generic thin-film	FB divider lower (V_OUT set)
R3	10 k ohm	0402	Generic	EN pullup
R4	10 k ohm	0402	Generic	BOOT_SEL (GPIO09) pullup
R5	5.1 k ohm	0402	Generic (x2, CC1/CC2)	USB-C CC pulldowns
JP1	2-pin solder jumper	SMD pad pair	Generic	120 ohm CAN termination (default open)
R_TERM	120 ohm 1%	0603	Generic	CAN termination resistor (footprint only)
J1	J1962 OBD-II connector	Through-hole / cable mount	TE / generic OBD-II shell	Vehicle bus interface
J2	USB-C receptacle	SMD, 16-pin	GCT / Amphenol USB4085/4110	Flashing / bench power
SW1	Tactile switch	SMD 6mm	Generic	Manual bootloader entry
LED1	Status LED (red/green)	0603	Generic	Active-low status indicator

4.2 Board 2 — Cockpit Display Node BOM

Ref	Value / Part	Package	Recommended P/N	Function
U1	ESP32-S3-WROOM-1-N8R8	SMD Module	Espressif ESP32-S3-WROOM-1-N8R8	Dual-core MCU / PSRAM / RF
U2	GC9A01 display module	FPC + round glass	Waveshare 1.28" round LCD	240x240 SPI TFT controller/panel

Ref	Value / Part	Package	Recommended P/N	Function
U3	CST816S	COF on FPC (module)	Hynitron CST816S	Capacitive touch controller
U4	TP4056	SOP-8 (ESOP8)	Nanjing Top Power TP4056	Single-cell LiPo linear charger
U5	ME6211C33M5G	SOT-23-5	Microne / Wuxi ME6211C33M5G	3.3V LDO regulator
Q1	AO3401A	SOT-23	Alpha & Omega AO3401A	P-channel power-path MOSFET
D1	SS14 / BAT54C	SMA / SOT-23	Diodes Inc. SS14	USB-to-rail power-path diode
D2	SMAJ5.0A	SMA / DO-214AC	Littelfuse SMAJ5.0A	VBUS OV clamp
U6	USBL6-2SC6	SOT-23-6	STMicroelectronics USBL6-2SC6	USB D+/D- ESD array
F1	PTC resettable fuse	0805	Bourns MF-USMF / Littelfuse	500mA hold / 1A trip
BT1	LiPo cell, 3.7V	Pouch, JST-PH connector	500-1000 mAh generic LiPo	Battery energy storage
R1	100 k ohm	0402	Generic	P-FET gate pulldown
R2	R_PROG (charge current set)	0402	Generic 1%, ~1.2k for 400mA	TP4056 charge current program
R3,R4	100 k ohm / 100 k ohm	0402	Generic 1%	Battery voltage ADC divider
J1	USB-C receptacle	SMD, 16-pin	GCT / Amphenol USB4085/4110	Power / flashing
J2	JST-PH 2-pin	Through-hole SMD	JST PH series	Battery connector
SW1	Tactile switch	SMD 6mm	Generic	Manual bootloader entry

Table 4.1/4.2 — Consolidated BOM. LCSC/DigiKey stocking numbers should be confirmed against current distributor inventory at fabrication time, as second-source equivalents (noted in §2 and §3) are acceptable substitutes for every active IC.

5. Manufacturing & PCB Design Guidelines

5.1 Stackup Specification

Layer	Function	Spec
Layer 1 (Top)	Signal + component placement	1 oz (35 μ m) copper
Core dielectric	FR-4, $T_g \geq 150^\circ\text{C}$	≈ 1.51 mm (standard 2-layer core)
Layer 2 (Bottom)	Ground pour + secondary routing	1 oz (35 μ m) copper

Table 5.1 — 2-layer stackup used for both Board 1 and Board 2. Overall finished thickness targets the standard 1.6 mm panel spec for mechanical compatibility with generic enclosures/connectors.

Both boards are fabricated as 2-layer designs: a top signal/component layer and a bottom layer used predominantly as a continuous ground pour, broken only where necessary for short jumper traces. 1 oz copper weight is sufficient for the current levels present on both boards (peak well under 2 A on the heaviest buck output trace) while keeping fabrication cost and lead time minimal for a capstone-scale production run.

5.2 High-Frequency Switching Loop Routing (Buck Converter)

The TPS54302's switch node (SW pin, inductor, and high-side/low-side FET commutation loop) is the dominant EMI source on Board 1 and must be treated as a high-di/dt loop rather than an ordinary signal net:

- Place the input bulk capacitor (C1) and HF bypass capacitor (C2) directly adjacent to the IC's VIN and GND pins, minimizing the loop area between VIN, the capacitor, and the IC ground return — this loop, not the output inductor loop, is the primary radiator.
- Route the SW node (IC pin to inductor) as a short, wide trace with no other signals routed underneath on the opposite layer, since it carries the full switching-frequency square wave at rail voltage amplitude.
- Place the output filter capacitors (C3, C4) immediately at the inductor's output pin, with their ground return tied directly into the same ground pour node as the IC's power-stage ground pin — not routed around through the digital ground of the ESP32-C3.
- Keep the feedback trace (R1/R2 midpoint to FB pin) away from the SW node and inductor; route it on the quiet side of the layout and, where the trace must cross the SW net, cross perpendicular rather than parallel to minimize capacitive/inductive coupling into the feedback path.

5.3 Differential CAN Trace Rules

CAN_H and CAN_L are routed from the J1962 connector, through the NUP2105L ESD array, into the SN65HVD230 as a matched differential pair:

- Target differential impedance: 120 Ω (matching the ISO 11898-2 bus specification), achieved on a 2-layer FR-4 stackup with an edge-coupled microstrip geometry — trace width and spacing should be derived from a field-solver calculation against the actual fabricator's dielectric height and copper weight rather than assumed.
- Length matching: intra-pair skew held within 50 mil (≈ 1.27 mm) end-to-end, with any necessary length compensation added as a serpentine on the shorter of the two nets immediately after the point of mismatch, not concentrated at the connector or IC end.
- Route the pair over a continuous, unbroken ground reference (no plane splits or via stitching gaps beneath the pair) for its entire length to maintain a consistent return path and avoid impedance discontinuities.

- Keep at least 3x the trace-to-trace spacing as clearance between the CAN pair and any other switching net, particularly the buck converter's SW node and the ESP32-C3's crystal traces.

5.4 Antenna RF Ground-Plane Clearance

Both the ESP32-C3-WROOM-02 and ESP32-S3-WROOM-1 modules integrate a PCB trace/chip antenna along one edge of the module footprint. On both boards:

- No copper, ground pour, silkscreen, or solder mask cutout artifacts are placed on any layer directly beneath the vendor-specified antenna keepout region — including the bottom-layer ground pour, which must be relieved under the antenna, not just the top layer.
- The module is placed at the physical edge of the PCB with the antenna pointing outward, and no metal enclosure wall, connector shell, or battery pouch (on Board 2) is placed within the module vendor's minimum clearance distance of the antenna edge.
- Long conductive traces (including the CAN differential pair on Board 1 and the SPI display bus on Board 2) are routed away from the antenna keepout zone rather than skirting its boundary, since parallel routing close to the keepout edge can still detune the antenna even outside the strict no-copper region.

5.5 Board 2 — Battery and Charge Path Layout Notes

The TP4056's thermal pad and R_{PROG} sense resistor trace are kept short and thermally isolated from the ME6211C33 LDO's dissipation path to avoid cross-heating that would otherwise skew the programmed charge current. The AO3401A footprint orientation is verified against the schematic's body-diode polarity before release to fabrication: a mirrored footprint silently reverses the power-path behavior described in §3.4, producing a board that appears to power up normally on the bench (USB present) but fails immediately upon USB disconnect — a failure mode that is easy to miss without deliberately testing the unplug transition during bring-up.